

Project Proposal — Smart Parking System

Advanced Embedded Systems Lab

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Summary & motivation We propose to build a Smart Parking System: an IoT application that detects in real time whether individual parking spaces are free or occupied and guides drivers to the nearest available space. Searching for parking in busy areas wastes time and fuel and adds to congestion, because drivers cannot see which spaces are free without physically searching. Each space is monitored by a wireless sensor node that reports to a central gateway, which makes the data available through a live dashboard and on-site indicator lights.

Objectives Detect occupancy of individual spaces reliably and in real time; transmit each space's status wirelessly to a central gateway; aggregate the data and count free spaces; display live availability on a dashboard and via on-site LEDs; and guide drivers to the nearest free space. The prototype will instrument three to four spaces, with a design that scales to a full lot.

Concept Ultrasonic sensors detect whether each space is occupied. The sensors connect to an ESP32 node, which evaluates the readings, drives a status LED per space (green = free, red = occupied), and publishes the data over Wi-Fi using MQTT to a Raspberry Pi gateway. A Python backend on the Pi maintains the state of all spaces, stores events in a database, and serves a web dashboard.

Technologies: ESP32 (C/C++), HC-SR04 ultrasonic sensors, Wi-Fi + MQTT, Raspberry Pi, Python (Mosquitto broker, Flask), SQLite, HTML/CSS/JS; code on GitHub.

Required hardware (request from lab)

- ESP32 development board (1–2) — sensor node
- HC-SR04 ultrasonic sensors (3–4) — occupancy detection
- RGB or red/green LEDs (3–4) + resistors — on-site status indication
- Raspberry Pi (Pi 4 or 3B+) + microSD card + power supply — gateway, broker, dashboard
- Breadboard, jumper wires, USB cable — prototyping and wiring

Optional: servo motor + barrier (entry gate), IR sensors, small OLED display.

Deliverables: working prototype (3–4 spaces), live dashboard with guidance, final documentation (~8 pages), source code on GitHub, and a demonstration.

We kindly ask for your approval of this project and for the hardware listed above.